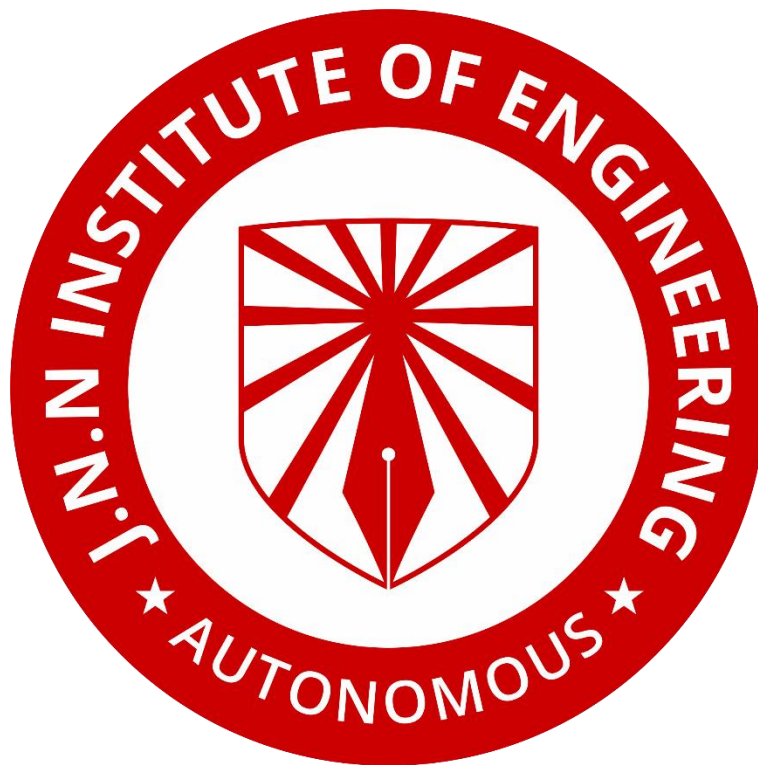


J.N.N Institute of Engineering
AUTONOMOUS

90, Ushaa Garden, Kannigaipair,
Chennai Periyapalayam Highway,
Tamil Nadu 601102

QUESTION BANK

COMMON TO CSE, CYBER SECURITY, AI&DS



III SEMESTER

25BST301– DISCRETE MATHEMATICS

Regulation–2025

Academic Year 2026-2027 (Odd Semester)



J.N.N INSTITUTE OF ENGINEERING

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90, Ushaa Garden, Kannigaipair, Chennai - Periyapalayam Highway,
Tiruvallur Dist, Tamil Nadu - 601 102.

QUESTION BANK

SUBJECT NAME: 25BST301- DISCRETE MATHEMATICS

Sem / Year: III / II

Unit-1 LOGIC AND PROOFS			
Propositional logic – Propositional equivalences - Predicates and quantifiers – Nested quantifiers – Rules of inference - Introduction to proofs – Proof methods and strategy.			
<u>PART-A</u>			
Q.No.	Questions	BT Level	Competence
1	Prove that $p \vee \neg p$ is a tautology.	BT3	Apply
2	Find the contra positive of “If it rains today then I buy an umbrella”.	BT4	Analyse
3	Find the truth table for $\neg p \rightarrow (\neg q \vee p)$	BT4	Analyse
4	Write the symbolic representation of “if it’s not hot today, then I buy and AC”.	BT1	Remember
5	Define conjunction and give an example.	BT1	Remember
6	Show that p and $\neg \neg p$ are equivalent.	BT3	Apply
7	Find the negation of $x, y p(x, y)$.	BT4	Analyse
8	Find the contra positive of implication “If it’s Sunday then it is a holiday”.	BT4	Analyse
9	Define disjunction and give an example.	BT1	Remember
10	How many are needed for the truth table for the formula $(p \wedge \neg q) \leftrightarrow (\neg r \wedge q)$.	BT2	Understand
11	Define Normal forms and also write its types.	BT1	Remember

12	Give the symbolic form of “Some men are optimistic”.	BT1	Remember
13	Write the symbolic representation if “If it’s rains today then I buy an umbrella”.	BT1	Remember
14	If 7 is a prime number, then that is not divided by 35. If 7 is divided by 35, concludes 7 is not prime number.	BT1	Remember
15	Find the truth table $p \rightarrow q$.	BT4	Analyse

PART-B

Q.No	Questions	Marks	BT Level	Competence
1	(i) Examine PDNF & PCNF of $(p \wedge q) \vee (\neg p \wedge r) \vee (q \wedge r)$ (ii) Test the validity of following statement, If Ram missing many classes due to illness, then he fails high school. If Ram fails high school, then he is uneducated. If Ram reads a lot of books, then uneducated. Therefore, Ram missing many classes due to illness and reads lot of books.	16	BT4	Analyse
2	(i) Test the validity of following statement “It is not sunny this afternoon and it is colder then yesterday we will go jogging only if it is sunny. If we don’t go jogging, then we will take a trip and if we take a trip then will be home sun set. Lead to the conclusion we will be home sunset”. (ii) Prove that $\neg r \vee p, r, q, p \rightarrow (q \rightarrow s) \Rightarrow s$	16	BT3	Apply
3	(i) Show that the premises, “One student in the class knows how to write a program in java and everyone who knows how to write in program in java can get high paying job. The conclusion, someone in this class can get a high paying job”. (ii) Show that the premises $p, p \rightarrow q, q \rightarrow \neg r, p \rightarrow r$ are inconsistent premises.	16	BT3	Apply
4	(i) Test the premises $P \rightarrow Q, Q \rightarrow \neg R, R, P \vee (R \vee S)$. (ii) Show that “t” is the valid conclusion from the premises $p \rightarrow q, q \rightarrow r, r \rightarrow s, \neg s, p \vee t$	16	BT3	Apply
5	(i) Obtain the truth table $(\neg p \rightarrow r) \wedge (q \rightarrow p) \wedge (q \rightarrow p)$. (ii) Show that $((p \vee q) \wedge \neg (\neg p \wedge (\neg q \vee \neg r))) \vee (\neg p \wedge \neg q) \vee (\neg p \wedge \neg r)$ is a tautology by using equivalences.	16	BT3	Apply
6	(i) Prove that $\sqrt{2}$ is an irrational number. (ii) Design a truth table $(\neg p \rightarrow r) \wedge (q \leftrightarrow p)$.	16	BT3	Apply

7	(i) Show that $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$ is a tautology. (ii) Evaluate the principal disjunctive normal form and principal conjunctive normal form $(p \rightarrow (q \wedge r)) \wedge (\neg p \rightarrow (\neg q \wedge \neg r))$	16	BT3	Apply
8	(i) Find the principal disjunctive normal form of $(\neg p \rightarrow q)$ and also find PCNF. (ii) Obtain the truth table $(\neg p \rightarrow r) \wedge (q \rightarrow p) \wedge (q \rightarrow p)$.	16	BT3	Apply
9	(i) Find the principal disjunctive normal form of $(p \wedge q) \vee (\neg p \wedge r) \vee (q \wedge r)$ and also find PCNF (ii) Prove that (a) $p \vee \neg q$ is a tautology. (b) $(p \rightarrow q) \wedge (p \rightarrow q)$ is contingency.	16	BT3	Apply
10	(i) Evaluate the principal disjunctive normal form and principal conjunctive normal form $(\neg p \wedge (\neg q \wedge \neg r)) \vee (q \wedge r) \vee (p \wedge r)$. (ii) Construct a truth table for $(p \leftrightarrow q) \leftrightarrow (r \leftrightarrow s)$	16	BT5	Evaluate

Unit-2 COMBINATORICS

Mathematical induction – Strong induction and well ordering – The basics of counting – The pigeonhole principle – Permutations and combinations – Recurrence relations – Solving linear recurrence relations – Generating functions – Inclusion and exclusion principle and its applications

PART-A

Q.No.	Questions	BT Level	Competence
1	If 7 colors are used to paint 57 bicycles. How many will have same colors?	BT2	Understand
2	How many different words are arranged in the word "INSTITUTION".	BT2	Understand
3	If $a_n = 2^n + 5(3)^n$, Find the first 3 terms.	BT4	Analyse
4	State inclusion and exclusion principle.	BT1	Remember
5	State generalized pigeon hole theorem.	BT1	Remember

6	How many different words are there in the word "MATHEMATICS"?	BT2	Understand
7	How many words will start with same letter among any group of 27 English letters.	BT2	Understand
8	How many different words are arranged in the word "ENGINEERING".	BT2	Understand
9	Find the first two terms of the sequence used by recurrence relations $a_n = 6a_{n-1}$, $a_0 = 2$.	BT4	Analyse
10	Find ${}^{30}C_{28}$.	BT4	Analyse
11	How many permutations are there in the word "MISSISSIPPI".	BT2	Understand
12	Find the permutations in the word "SUCCESS".	BT4	Analyse
13	How many integers between 1 and 100 that are divisible by 3 and not divisible by 7.	BT4	Analyse
14	Find the first two terms of the sequence used by recurrence relations $a_n = 6a_{n-1}$, $a_0 = 2$.	BT4	Analyse
15	How many positive integers are not exceeding 100 that is divisible by 5.	BT2	Understand

PART-B

Q.No	Questions	Marks	BT Level	Competence
1	Find the number of positive integers ≤ 1000 and not divisible any number of 3, 5, 7, 22?	16	BT3	Apply
2	(i) Find the number of integers between 1 and 500 that are divisible by integers 2, 3 and 5. (ii) Show that $1^2+2^2+3^2+\dots+n^2 = \frac{n(n+1)(2n+1)}{6}$ by mathematical induction method.	16	BT3	Apply
3	(i) Find the number of integers ≤ 500 that are not divisible by integers 2,3 and 5. (ii) Solve the recurrence relation for Fibonacci series.	16	BT3	Apply
4	(i) From a club consisting of 6 mens 15 and 7 women. In how many ways we select a committee of (a) Three men and Four women.	16	BT3	Apply

	(b) Four person which has at most one men. (c) Four person that has children of both six. (ii) There are 6 boys and 5 girls in a group. Find the number of ways 4 peoples can be selected. (a) They can be boy or girl. (b) 2 must be boy & 2 must be girl. (c) They are must be same sex.			
5	(i) Solve the recurrence relation $a_n = 8a_{n-1} - 16a_{n-2}$ with initial condition $a_2=16, a_3=80$ (ii) Solve the recurrence relation $a_n + 6a_{n-1} + 12a_{n-2} + 8a_{n-3} = 0$ with initial condition $a_0=1, a_1=2, a_2=4$.	16	BT3	Apply
6	(i) Solve $a_n + 6a_{n-1} + 12a_{n-2} + 8a_{n-3} = 0$ where $a_0=1, a_1=2, a_2=4$ by characteristic roots method. (ii) A committee of 5 to selected from 6 boys and 5 girls. Find the number of ways for selecting a committee if it consists at least one boy and one girl.	16	BT3	Apply
7	Find the number of positive integers ≤ 1000 and not divisible by 5, 7, 9 and 22.	16	BT3	Apply
8	(i) Solve the recurrence relation $s(k) = -3s(k-1) - 3s(k-2) - s(k-3)$ where $s(0)=0, s(1)=-2, s(2)=-1$ by characteristic roots method. (ii) A box contains 6 white balls and 5 red balls. Find the number of ways 4 balls can be selected. a. They can be white or red ball b. Two must white and two must red c. All must be same color.	16	BT3	Apply
9	(i) Prove by Mathematical induction method $1+2+3+\dots+n = \frac{n(n+1)}{2}, n \geq 1$. (ii) Solve the recurrence relation for $a_n - a_{n-1} - a_{n-2} = 0$, with initial value $a_0=1, a_1=1$.	16	BT3	Apply
10	(i) Prove that $\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} > \sqrt{n}$ for $n \geq 2$ using by mathematical induction method. (ii) Find the number of integers ≤ 250 that are not divisible by integers 2, 3 and 5	16	BT3	Apply

Unit-3 GRAPHS

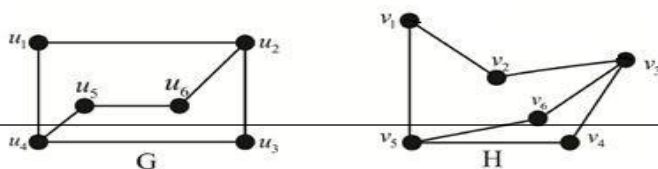
Graphs and graph models – Graph terminology and special types of graphs – Matrix representation of graphs and graph isomorphism – Connectivity – Euler and Hamilton paths.

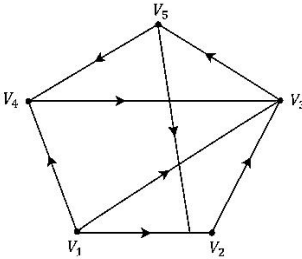
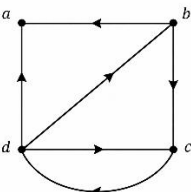
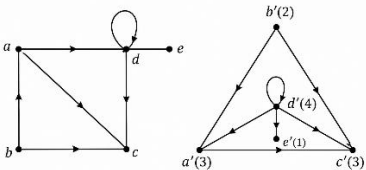
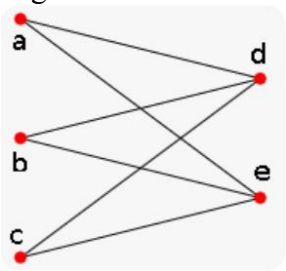
PART-A

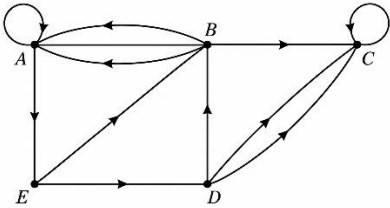
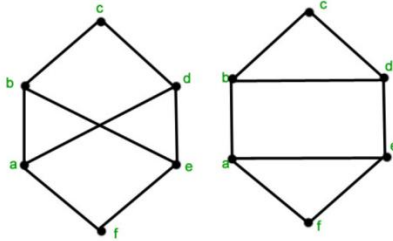
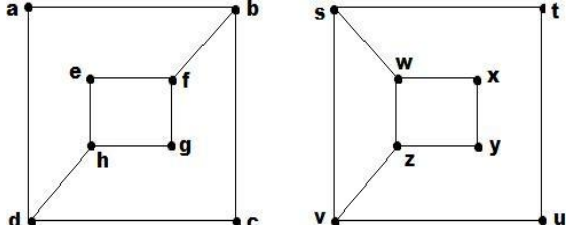
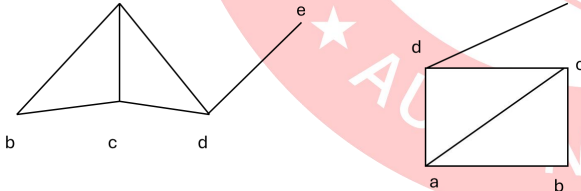
Q.No.	Questions	BT Level	Competence
1	Define Pseudo graph and give example.	BT1	Remember
2	Can a simple graph be existing with 15 vertices each of degree is 5?	BT4	Analyse
3	Define cyclic graph and draw the graph C_5 .	BT4	Analyse
4	If there is a graph corresponding to a following degree 4, 4, 4, 3, 2.	BT4	Analyse
5	Define complete graph and draw the graph K_5 .	BT4	Analyse
6	What is the number of edges in k regular graph.	BT4	Analyse
7	Define a complete bipartite graph of $K_{2,3}$.	BT1	Remember
8	Can be a regular graph be a complete graph.	BT4	Analyse
9	Define incidence matrix and give an example.	BT4	Analyse
10	What is the number of edges in complete graph, K_n .	BT4	Analyse
11	Define isomorphism of two graphs.	BT1	Remember
12	How many vertices and edges in W_5 .	BT4	Analyse
13	Define a pendent vertex in a graph.	BT1	Remember
14	Define adjacency matrix and give an example.	BT1	Remember
15	Define Euler graph and given an example.	BT1	Remember

Part B

Q.No	Questions	Marks	BT Level	Competence
1	(i) Prove that the number of odd degree vertices is always even. (ii) State and prove the fundamental theorem of graph theory.	16	BT3	Apply
2	Find the number of vertices, edges, degrees of vertices, maximum degree, minimum degree and also verify with handshaking theorem and check the graphs are isomorphism.	16	BT3	Apply



3	Find the number of vertices, edges, degrees of vertices, maximum degree, minimum degree and also verify with handshaking theorem.  (ii) Let $G = (V, E)$ be an undirected graph with e edges then the total $\deg(v) = 2e$.	16	BT5	Evaluate
4	Find the number of vertices, edges, degrees of vertices, maximum degree, minimum degree and also verify with handshaking theorem.  (ii) Exam whether the following pair of graphs are isomorphic or not. 	16	BT5	Evaluate
5	(i) Prove that a given connected graphs G is a Euler graphs if all vertices of G is even degree. (ii) Give an example for Eulerian but not Hamiltonian graph (iii) Hamiltonian but not Eulerian graph	16	BT3	Apply
6	Prove that, if G is a simple graph with n vertices and k components when the no. of edges at most $\frac{(n-k)(n-k+1)}{2}$	16	BT3	Apply
7	(i) For any simple bipartite graph G with edges and vertices prove that $e \leq v^2/4$ (ii) Find the number of vertices, edges, degrees of vertices and also verify with handshaking theorem, maximum degree and minimum degree. 	16	BT3	Apply

8	(i) Find the number of vertices, edges, degrees of vertices, maximum degree, minimum degree and also verify with handshaking theorem.  (ii) Check the given two graphs are isomorphism? 	16	BT3	Apply
9	Find the number of vertices, edges, degrees of vertices, maximum degree, minimum degree and also verify with handshaking theorem and check the graphs are isomorphism. 	16	BT3	Apply
10	Find the number of vertices, edges, degrees of vertices, maximum degree, minimum degree and also verify with handshaking theorem and check the graphs are isomorphism. 	16	BT3	Apply

UNIT IV ALGEBRAIC STRUCTURES, LATTICES AND BOOLEAN ALGEBRA

Algebraic systems – Semi groups and monoids - Groups – Subgroups – Homomorphism's – Normal subgroup and cosets – Lagrange's theorem – Definitions and examples of Rings and Fields- Partial ordering – Posets – Lattices as posets – Properties of lattices - Lattices as algebraic systems – Sub lattices – Direct product and homomorphism – Some special lattices – Boolean algebra – Sub Boolean Algebra – Boolean Homomorphism.

PART-A

	Questions	BT Level	Competence
1	If $f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 4 & 5 \end{pmatrix}$ Find f^{-1} .	BT3	Evaluate
2	Define normal sub group.	BT1	Remember
3	When is a $(G, *)$ is abelian.	BT2	Understand
4	Define Homomorphism and give an example.	BT1	Remember
5	State any two properties of group.	BT1	Remember
6	If $g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 3 & 4 & 5 \end{pmatrix}$ Find g^{-1} .	BT3	Evaluate
7	Show that inverse element is unique in $(G, *)$.	BT3	Apply
8	Find the cosets of the subgroup $H = \{[0]\}$ of the group $[Z_2, +]$.	BT1	Remember
9	In Boolean algebra with 5 elements is possible? Justify your answer.	BT2	Understand
10	Prove that $a.b' + a'.b = 0$	BT5	Evaluate
11	Write the properties of Boolean algebra	BT1	Remember
12	Give an example of a distributive lattice but not complemented.	BT2	Understand
13	Define equivalence relation.	BT1	Remember
14	If $X = \{1, 2, 3, 4, 5, 6\}$ and R be a relation defined as $(x,y) \in R$ iff $x-y$ is divisible by 3. Find the elements of the relation R .	BT1	Remember
15	Give an example of a lattice that is not complemented.	BT2	Understand

Part B

Q.No	Questions	Marks	BT Level	Competence
1	(i) State and prove Lagrange's theorem. (ii) Find the Cosets of the sub group $H = \{[0],[3]\}$ of the group $(Z_6, +)$.	16	BT1	Remember

2	(i) If any group $\{G, *\}$ prove that $(a*b)^{-1} = b^{-1} * a^{-1}$, for every a, b are belongs to G. (ii) Prove that $G = \{1, w, w^2\}$ is an Abelian group by using binary multiplication.	16	BT3	Apply
3	(i) Show that Z_8 is an abelian under the addition modulo 8. (ii) Show that the set of all real numbers except -1 is an abelian group under operation $*$ defined by $a*b = a + b + ab$.	16	BT3	Apply
4	(i) Show that the set of all non-zero real numbers is an abelian group under operation $*$ defined by $a*b = ab/2$ (ii) In $(G, *)$ prove that (i) Identity elements is unique (ii) Inverse element is unique	16	BT3	Apply
5	(i) In Show that Z_6 is an abelian under the addition modulo 6. (ii) a group $(G, *)$ the cancellation law is true, for every a, b, c is belonging to G. prove that (a) $a*b = a*c$ then $b=c$. (b) $b*a = c*a$ then $b=c$.	16	BT3	Apply
6	(i) Prove that every chain is a distributive lattice (ii) Consider $x = \{2, 3, 6, 12, 24, 36\}$ and $R = \{ (a,b)/a/b \}$. Draw the Hasse diagram.	16	BT3	Apply
7	(i) Prove that a chain with three or more elements is not complement. (ii) Prove that every distributive lattice is modular.	16	BT3	Apply
8	(i) State and prove Demorgan's law. (ii) Consider $x = \{2, 4, 5, 10, 20, 25, 40, 50\}$ and $R = \{ (a,b)/a/b \}$. Find the LCB and GLB.	16	BT1	Remember
9	(i) Prove that every distributive lattice is modular but not conversely. (ii) Prove that every chain is modular.	16	BT3	Apply
10	(i) Consider $x = \{2, 3, 6, 12, 24, 36\}$ and $R = \{ (a,b)/a/b \}$. Draw the Hasse diagram and also find glb of (2, 3) and (24, 36) (ii) Show that in a distributive and complemented lattice satisfied demorgan's laws.	16	BT3	Apply